

ELAMARAN SUBRAMANIAN

+91 9490102695 | subramanianelamaran@gmail.com
github.com/elamaran-2004 | linkedin.com/in/elamaran-subramanian-330a42329

EDUCATION

SRM Institute of Science and Technology, Kattankulathur B.Tech, Computer Science & Engineering CGPA: 7.58/10	May 2026
SBOA Matriculation Higher Secondary School, Chennai Class XII-82.5%	May 2022
DAV High School Class X-97%	March 2020

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks
Web Technologies: HTML5, CSS3, JavaScript, React.js, REST APIs
Databases: MySQL
Cloud Technologies: Amazon Web Services (AWS)
Developer Tools: Git, GitHub, IntelliJ IDEA, Visual Studio Code, Postman
Enterprise Technologies: REST APIs, Authentication, API Security, Redis, System Design Basics
Artificial Intelligence: Machine Learning, Generative AI, Large Language Models (LLMs)

INTERNSHIP EXPERIENCE

AI- Intern Retech Solutions	May 2025
- Completed a 15-day industry-oriented AI internship; executed a hands-on project on Fraud Detection in Financial Transactions using supervised machine learning techniques to classify fraudulent vs. legitimate transaction patterns. - Gained applied exposure to ML model development, feature engineering, and data-driven anomaly detection workflows relevant to security analytics.	

PROJECT EXPERIENCE

API Rate Limiting & Access Control System using JWT and Redis

Technologies: Spring Boot, Java, JWT, Redis, REST APIs, Postman, Maven

- Developed a secure REST API gateway with JWT authentication, role-based authorization, and Redis-backed rate limiting for scalable request management.
- Implemented a Redis-backed rate-limiting engine to monitor, throttle, and control inbound API request traffic in real time, mitigating abuse vectors such as brute-force and denial-of-service attempts.
- Built a token validation pipeline to verify JWT integrity, expiry, and role claims on every API call, ensuring only authorized principals gain access to protected resources.
- Conducted end-to-end security testing of authentication flows, rate-limit thresholds, error handling, and response integrity using Postman.

Medical Attention Transformer using Computer Vision for Multi-Organ Tumor Segmentation

Technologies: Python, Deep Learning, Computer Vision, Vision Transformers (ViT), UNETR, PyTorch, Medical Image Analysis.

- Developed a transformer-based deep learning framework for automated multi-organ tumor segmentation from MRI and CT datasets
- Implemented Vision Transformer (ViT) and UNETR-based architectures to extract spatial features and improve segmentation accuracy.
- Utilized BraTS and liver imaging datasets with preprocessing techniques including normalization and data augmentation.
- Evaluated model performance using segmentation metrics and optimized training through hyperparameter tuning and learning rate scheduling.

LLM-Enhanced Spatiotemporal Prediction for EV Charging Load Management

Technologies: React.js, Python, LSTM, Google Gemini API, Machine Learning, Data Analytics, Data Visualization

- Developed a spatiotemporal LSTM forecasting model to predict EV charging load demand, to forecast charging demand and optimize charging resource allocation across multiple locations.
- Integrated Google Gemini LLM to deliver AI-driven optimization recommendations and decision-support insights based on predictive analytics outputs.
- Built an interactive React.js dashboard featuring real-time forecasts, geographic heatmaps, threshold-triggered alerts, and optimization summaries to support operational decision-making.

CERTIFICATIONS

- Google AI Essentials – Google
- Oracle Certified Foundations Associate – Java
- Meta Front-End Developer Professional Certificate
- HackerRank – Software Engineer Certificate
- SQL (Intermediate) – Certificate of Accomplishment, HackerRank
- AWS Academy Graduate – Machine Learning Foundations

HACKATHONS & TECHNICAL PARTICIPATION

- IDEATHON 2024 – Collaborated in a team to develop and present an innovative technology-based product concept addressing a real-world problem.
- ReGenHack'25 Hackathon – Competed in a regenerative technology hackathon, collaborating in a team to design and prototype a solution under time-constrained conditions.

PROFESSIONAL MEMBERSHIP

- Member, Institution of Engineers (India) (IEI) – Since 2023